DAS-MIL: Distilling Across Scales for MIL Classification of Histological WSIs

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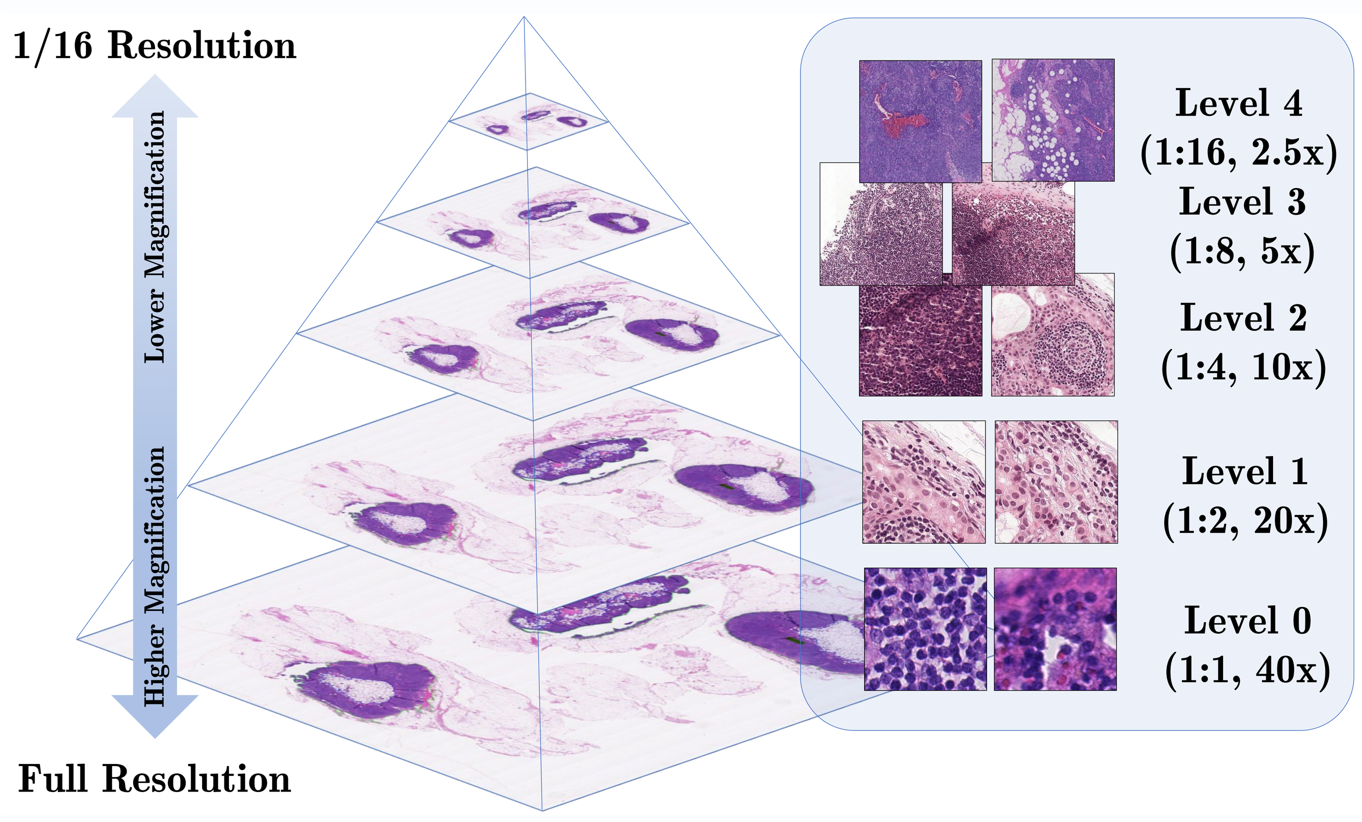
SCAN ME

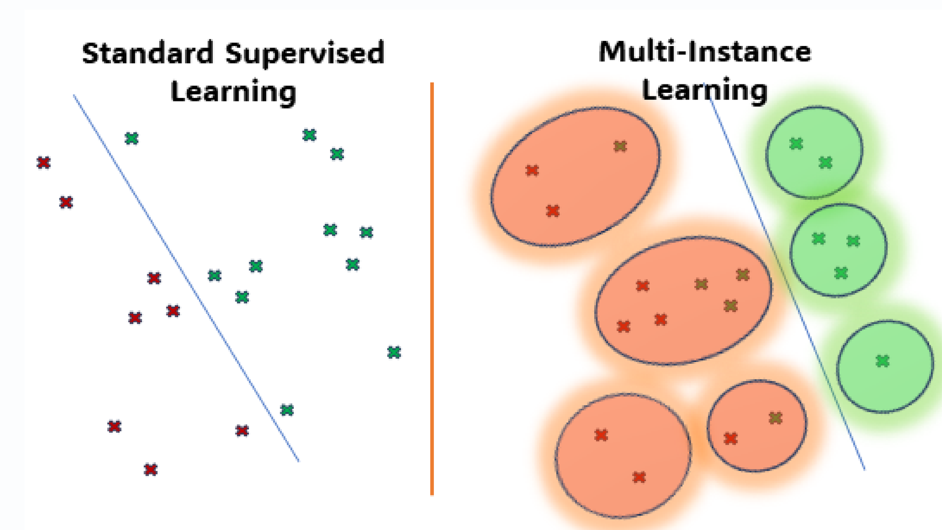
Classification of WSIs



Modern microscopes allow the digitization of conventional glass slides into gigapixel Whole-slide Images (**WSIs**).

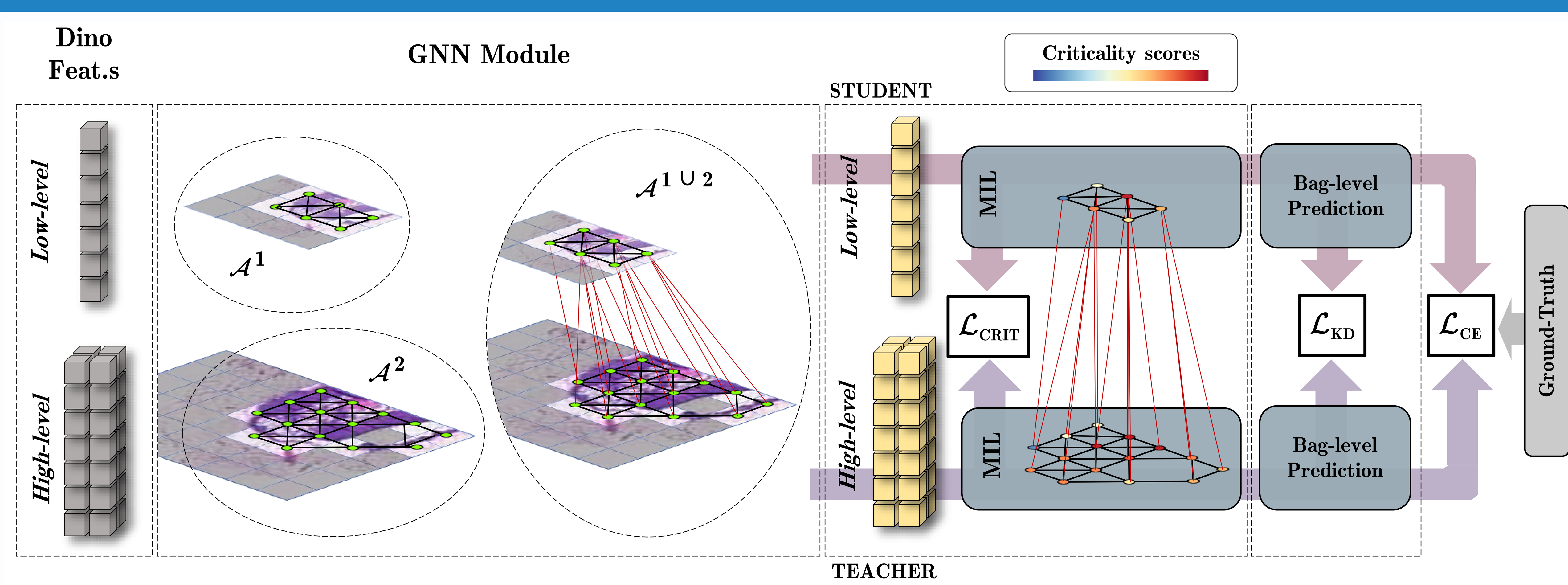
Multiple Instance Learning (MIL)

Annotating WSIs is **expensive** and **time-consuming**, and labels are usually provided at slide or patient level. Feeding modern neural networks with the entire gigapixel image is **not feasible**.



A **MIL** approach considers the slide as a bag composed of many patches, called instances. The bag is positive only if it contains at least one positive instance.

Distilling Across Scales (DAS-MIL)



Hierarchical Graph Context: an efficient hierarchical structure allows to share information along neighbors and scales.

MIL for each scale: an attention mechanism solves the task at each resolution.

Bag Self-Distillation

$$\mathcal{L}_{KD} = \tau^2\ \mathsf{KL}(\mathrm{softmax}(\frac{y^{\mathrm{BAG}}_{scale1}}{\tau}) \parallel \mathrm{softmax}(\frac{y^{\mathrm{BAG}}_{scale2}}{\tau}))$$

Patch Self-Distillation

$$\mathcal{L}_{CRIT} = \|\mathrm{z}_{scale1} - \mathrm{GraphPooling}(\mathrm{z}_{scale2})\|_2^2$$

Loss

$$\min_{\theta}\ (1-\lambda)\mathcal{L}_{CE}(y^{\mathrm{BAG}}_{scale2}) + \mathcal{L}_{CE}(y^{\mathrm{BAG}}_{scale1}) + \lambda\mathcal{L}_{KD} + \beta\mathcal{L}_{CRIT}$$

where β, λ, τ are hyperparameters.

Results

		Camelyon16		TCGA Lung	
	Method	Accuracy	AUC	Accuracy	AUC
Single	Mean-pooling	0.723	0.672	0.823	0.905
Single	Trans-MIL [2]	0.883	0.942	0.881	0.948
Single	DTFD (AFS) [3]	0.908	0.946	0.891	0.951
Single	DSMIL [1]	0.915	0.952	0.888	0.951
Multi	HIPT [4]	0.890	0.951	0.890	0.950
Multi	DSMIL-LC [1]	0.909	0.955	0.913	0.964
Multi	**DAS-MIL** (ours)	**0.945**	**0.973**	**0.925**	**0.965**

Our model outperforms previous SOTAs

Hyper Parameter Selection

λ	20×	10×	β	20×	10×
1.0	**0.973**	**0.974**	1.5	0.964	0.968
0.8	0.967	0.966	1.2	0.970	0.964
0.5	0.968	0.932	1.0	**0.973**	**0.974**
0.3	0.962	0.965	0.8	0.962	0.965
0.0	0.955	0.903	0.6	0.951	0.953

τ	Accuracy		AUC	
	20×	10×	20×	10×
$\tau = 1$	0.883	0.962	0.906	0.957
$\tau = 1.5$	**0.945**	**0.945**	**0.973**	**0.974**
$\tau = 2$	0.906	0.914	0.962	0.963
$\tau = 2.5$	0.922	0.914	0.951	0.952

The beneficial effect of self-distillation is demonstrated when λ and β are high and τ is low.

[1] Li, B., et al.: Dual-stream Multiple Instance Learning Network for Whole Slide Image Classification with Self-supervised Contrastive Learning. CVPR2021

[2] Shao, et al.: TransMIL: Trans-former based Correlated Multiple Instance Learning for Whole Slide Image Clas-sification. NIPS 2021

[3] Zhang, et al.: DTFD-MIL: Double-Tier Feature Distillation Multiple Instance Learning for Histopathology Whole Slide Image Classification. CVPR 2022

[4] Chen, et al.: Scaling Vision Transformers to Gigapixel Images via Hierarchical Self-Supervised Learning. CVPR 2022